

STACK publications

This document contains publications relating to the STACK computer aided assessment system.
For more information about STACK please see <https://stack-assessment.org/>
The original BiBTeX entries are available from:

https://github.com/mathsmoodle-qtype_stack/tree/master/doc/content/stack.bib

1 Suggestions of where to start

- Kinnear, G., Jones, I., Sangwin, C., Alarfaj, M., Davies, B., Fearn, S., Foster, C., Heck, A., Henderson, K., Hunt, T., Iannone, P., Kontorovich, I., Larson, N., Lowe, T., Meyer, J. C., O'Shea, A., Rowlett, P., Sikurajapathi, I., & Wong, T. (2024). A collaboratively-derived research agenda for E-assessment in undergraduate mathematics. *International Journal of Research in Undergraduate Mathematics Education*, 10(1), 201–231. <https://doi.org/10.1007/s40753-022-00189-6>
- O'Hagan, S., Sangwin, C. J., & Zerva, K. (2022). *A report on the use of the STACK service in Edinburgh, 2021–22* (tech. rep.). University of Edinburgh. Edinburgh, UK.
- Sangwin, C. J. (2013). *Computer aided assessment of mathematics*. Oxford University Press.
- Sangwin, C. J., & Bickerton, R. (2023). Practical online assessment of mathematical proof. *International Journal of Mathematical Education in Science and Technology*, 53(10). <https://doi.org/10.1080/0020739X.2021.1896813>
- Sangwin, C. J., & Jones, I. (2017). Asymmetry in student achievement on multiple choice and constructed response items in reversible mathematics processes. *Educational Studies in Mathematics*, 94, 205–222. <https://doi.org/10.1007/s10649-016-9725-4>

2 Books

- Nakamura, Y. (2010). *The stack e-learning and assessment system for mathematics, science and engineering education through moodle* [(In Japanese)]. Tokyo Denki University Press.
- Sangwin, C. J. (2013). *Computer aided assessment of mathematics*. Oxford University Press.

3 Papers on specific issues

- Nakamura, Y., & Takahara, T. (2016). Development of a math input interface with flick operation for mobile devices. *12th International Conference on Mobile Learning, 9–11 April, Vilamoura, Algarve, Portugal*.
- Sangwin, C. J. (2015). Inequalities, assessment and computer algebra. *International Journal of Mathematical Education in Science and Technology*, 46(1), 76–93. <https://doi.org/10.1080/0020739X.2014.941424>

- Sangwin, C. J. (2019). Proof technology in mathematics research and teaching. Springer International.
- Sangwin, C. J., & Harjula, M. (2017). Online assessment of dimensional numerical answers using STACK in science. *European Journal of Physics*.
- Sangwin, C. J., & Ramsden, P. (2007). Linear syntax for communicating elementary mathematics. *Journal of Symbolic Computation*, 42(9), 902–934. <https://doi.org/10.1016/j.jsc.2007.07.002>

4 STACK and Examinations

- Sangwin, C. J. (2018). High stakes automatic assessments: Developing an online linear algebra examination. *Proceedings of 11th Conference on Intelligent Computer Mathematics*.
- Sangwin, C. J. (2019). Developing and evaluating an online linear algebra examination for university mathematics. *Proceedings of CERME 11*, (TWG 21: Mssessment in Mathematics Education).
- Sangwin, C. J., & Köcher, N. (2016). Automation of mathematics examinations. *Computers and Education*, 94, 215–227. <https://doi.org/10.1016/j.compedu.2015.11.014>

5 Theses

- Badger, M. (2013). *Problem-solving in undergraduate mathematics and computer aided assessment* (PhD). University of Birmingham.
- Cervál-Peña, E. R. (2008). *Automated computer-aided formative assessment with ordinary differential equations* (Master's thesis). University of Birmingham.
- Harjula, M. (2008). *Mathematics exercise system with automatic assessment* (Master's thesis). Helsinki University of Technology.
- Majander, H. (2010). *Tietokoneavusteinen arviointi kurssilla diskreetin matematiikan perusteet* (Master's thesis). University of Helsinki.
- Majander, H., & Rasila, A. (2011). Tutkimus suuntaamassa 2010-luvun matemaattisten aineiden opetusta. Tampereen yliopistopaino Oy - Juvenes Print.
- Ruokokoski, J. (2009). *Automatic assessment in university-level mathematics* (Master's thesis). Helsinki University of Technology.
- Tanskanen, H. (2010). *Dynaamista geometriaa moodle-ympäristöön - stack- ja jsxgraph-järjestelmien testaamista monimuotoisten kysymysten laatimiseksi* (Master's thesis). University of Eastern Finland.

6 Research and conference papers

- Akveld, M., & Kinnear, G. (2024). Improving mathematics diagnostic tests using item analysis. *International Journal of Mathematical Education in Science and Technology*, 55(10), 2478–2505. <https://doi.org/10.1080/0020739X.2023.2167132>

- Badger, M., & Sangwin, C. (2011). My equations are the same as yours!: Computer aided assessment using a Gröbner basis approach. In A. A. Juan, M. A. Huertas, & C. Steegmann (Eds.), *Teaching mathematics online: Emergent technologies and methodologies* (pp. 259–273). IGI Global. <https://doi.org/10.4018/978-1-60960-875-0.ch013>
- Barbas, H., & Schramm, T. (2016). The Hamburg online math test MINTFIT for prospective students of STEM degree programmes. *Proceedings of SEFI, Tampere, Finland*.
- Bradford, R., Davenport, J. H., & Sangwin, C. J. (2009). A comparison of equality in computer algebra and correctness in mathematical pedagogy. *Proceedings of Calculemus*, (5625), 75–89. https://doi.org/10.1007/978-3-642-02614-0_11
- Bradford, R., Davenport, J. H., & Sangwin, C. J. (2010). A comparison of equality in computer algebra and correctness in mathematical pedagogy (ii). *The International Journal for Technology in Mathematics Education*, 17(2), 93–98.
- Butcher, P., Sangwin, C., & Hunt, T. (2013). Embedding and enhancing eassessment in the leading open source VLE. *Proceedings of the Higher Education Academy Conference, Birmingham*.
- Cameron, R. A. (2024). Designing a blended delivery foundation mathematics course: Targeting self-efficacy, skill development and social connectedness. *MSOR Connections*, 22(1), 29–41. <https://doi.org/10.21100/msor.v22i1.1451>
- Cameron, R. A. (2025). ‘The green tick’: How instant feedback shaped self-efficacy in a first-year mathematics course. *International Journal of Mathematical Education in Science and Technology*, 1–17. <https://doi.org/10.1080/0020739X.2025.2574947>
- Chelleri, E., Doz, D., Juma, Z. O., d’Assise Dogbalou, M. W., & Lewański, D. (2025). Assessing student perspectives on STACK implementation in university mathematics courses. *International Journal of Mathematical Education in Science and Technology*, 1–32. <https://doi.org/10.1080/0020739X.2025.2563115>
- Doz, D., d’Assise Dogbalou, M. W., Juma, Z. O., & Lewański, D. (2026). Latent class analysis of student attitudes and performance in online mathematics assessments: A Technology Acceptance Model approach. *International Journal of Mathematical Education in Science and Technology*, 57(5), 944–966. <https://doi.org/10.1080/0020739X.2025.2502859>
- Erskine, G., & Mestel, B. (2018). Developing STACK practice questions for the Mathematics Masters Programme at the OU. *MSOR Connections*, 17(1). <https://doi.org/10.21100/msor.v17i1.896>
- Gratwick, R., Kinnear, G., & Wood, A. K. (2020). An online course promoting wider access to university mathematics. *Proceedings of the British Society for Research into Learning Mathematics*, 40(1), 1–6. <https://bsrlm.org.uk/wp-content/uploads/2020/05/BSRLM-CP-40-1-04.pdf>
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- Iannone, P., Lemmo, A., & Kinnear, G. (2026). The transfer of assessment from pen and paper to e-assessment: Two case studies from one theoretical perspective. In E. Geraniou, C. Crisan, & M. Mavrikis (Eds.), *Digital Technology and Artificial Intelligence in Mathematics Education Assessment*. Routledge. <https://doi.org/10.4324/9781003533764-2>

- Keady, G., Fitz-Gerald, G., Gamble, G., & Sangwin, C. (2006). Computer-aided assessment in mathematical sciences. *UniServe Science Symposium, University of Sydney*, 69–73.
- Kinnear, G. (2024). Comparing example generation with classification in the learning of new mathematics concepts. *Research in Mathematics Education*, 26(1), 109–132. <https://doi.org/10.1080/14794802.2022.2152086>
- Kinnear, G., & Iannone, P. (2024). Students' views of e-assessment feedback in undergraduate mathematics. In T. Evans, O. Marmur, J. Hunter, G. Leach, & J. Jhagroo (Eds.), *Proceedings of the 47th Conference of the International Group for the Psychology of Mathematics Education* (pp. 137–144). PME. <https://doi.org/10.5281/zenodo.15348008>
- Kinnear, G., Iannone, P., & Davies, B. (2023). Insights about functions from example-generation tasks: Combining e-assessment and written responses. In P. Drijvers, C. Csapodi, H. Palmér, K. Gosztonyi, & E. Kónya (Eds.), *Proceedings of the Thirteenth Congress of the European Society for Research in Mathematics Education (CERME13)* (pp. 2399–2406). Alfréd Rényi Institute of Mathematics and ERME. <https://hal.science/hal-04406716>
- Kinnear, G., & Kontorovich, I. (2024). Using e-assessment for interactive example-generation tasks. In P. Iannone, F. Moons, C. Drüke-Noe, E. Geraniou, F. Morselli, K. Klingbeil, M. Veldhuis, & S. Olsher (Eds.), *Proceedings of FAME 1 – Feedback & Assessment in Mathematics Education (ETC 14)* (pp. 158–165). Utrecht University and ERME. <https://doi.org/10.5281/zenodo.14231455>
- Kinnear, G., Wood, A. K., & Gratwick, R. (2022). Designing and evaluating an online course to support transition to university mathematics. *International Journal of Mathematical Education in Science and Technology*, 53(1), 11–34. <https://doi.org/10.1080/0020739X.2021.1962554>
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- Lowe, T., & Hunt, T. (2026). A decade of computer-algebra-aided online assessment of mathematics at the UK Open University. *Teaching Mathematics and its Applications: An International Journal of the IMA*, 45(1), 1–18. <https://doi.org/10.1093/teamat/hraf021>
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- Nakamura, Y., Nakahara, T., & Akiyama, M. (2010). Teaching of differential equations using mathematics e-learning system STACK. *Proceedings of the IADIS International Conference e-Learning*.
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- Razeghpour, F. (2024). Conception of two informative tutorial feedback strategies for mathematical tasks with STACK. In P. Iannone, F. Moons, C. Druke-Noe, E. Geraniou, F. Morselli, K. Klingbeil, M. Veldhuis, & S. Olsher (Eds.), *Proceedings of FAME 1 – Feedback & Assessment in Mathematics Education (ETC 14)* (pp. 254–261). Utrecht University and ERME. <https://doi.org/10.5281/zenodo.14231455>
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- Shirai, S., & Fukui, T. (2015). Improving the math input method for matrices for use in linear algebra on STACK [(In Japanese)]. *IPSJ Transactions on Computers and Education*, 1(3), 22–29.
- Shirai, S., Nakamura, Y., & Fukui, T. (2015). An interactive math input method for computer aided assessment systems in mathematics [(In Japanese)]. *IPSJ Transactions on Computers and Education*, 1(3), 11–21.
- Taha, S., Kinnear, G., & Iannone, P. (2024). Perceptions of effective formative feedback: A comparative analysis between undergraduate students and mathematics lecturers. In P. Iannone, F. Moons, C. Drüke-Noe, E. Geraniou, F. Morselli, K. Klingbeil, M. Veldhuis, & S. Olsher (Eds.), *Proceedings of FAME 1 – Feedback & Assessment in Mathematics Education (ETC 14)* (pp. 331–332). Utrecht University and ERME. <https://doi.org/10.5281/zenodo.14231455>
- Wallace, C., & Fearn, S. (2026). Students writing STACK questions: E-assessment as the focus of a final-year project. *Teaching Mathematics and Its Applications*, 45(1), 124–138. <https://doi.org/10.1093/teamat/hrag006>
- Yoshitomi, K. (2018). Generation of abundant multi-choice or STACK type questions using cas for random assignments. In J. Davenport, M. Kauers, G. Labahn, & J. Urban (Eds.), *Proc. mathematical software — icms 2018* (pp. 492–497).

7 Proceedings of the first STACK conference, 2018

For copies of the proceedings of the first STACK conference see <https://zenodo.org/communities/stack>

- Altieri, M., Staupe, M., Paluch, D., & Schirmer, E. (2019). STACK trifft H5P – interaktives, audiovisuelles Feedback in STACK-Aufgaben basierend auf H5P. <https://doi.org/10.5281/zenodo.2564881>

- Bach, S. (2019). *STACK-Fragen zur Unterstützung der Selbstlernphasen in einem Studienvorbereitungsangebot für beruflich Qualifizierte*. Zenodo. <https://doi.org/10.5281/zenodo.2563803>
- Bird, A., & Rathmann, W. (2019). Ein Online-Trainingscenter zur Stärkung der Rechenfähigkeiten. *Contributions to the 1st International STACK conference 2018 in Fürth, Germany*. <https://doi.org/10.5281/zenodo.2563076>
- Eckhoff, M., & Landenfeld, K. (2019). GoMaxima - Eine performante Verwendung von STACK und Maxima in Lernmanagementsystemen. *Contributions to the 1st International STACK conference 2018 in Fürth, Germany*. <https://doi.org/10.5281/zenodo.2561190>
- Fath, J., Hansen, P., Scheicher, C., & Umbach, T. (2019). E-homework with individual feedback for large lectures. *Contributions to the 1st International STACK conference 2018 in Fürth, Germany*. <https://doi.org/10.5281/zenodo.2565860>
- Härterich, J. (2019). Using randomized quizzes in undergraduate linear algebra and multivariable calculus. *Contributions to the 1st International STACK conference 2018 in Fürth, Germany*. <https://doi.org/10.5281/zenodo.2582874>
- Kinnear, G. (2019). Delivering an online course using STACK. *Contributions to the 1st International STACK conference 2018 in Fürth, Germany*. <https://doi.org/10.5281/zenodo.2565969>
- Klischat, C., Becker, P., & Vasko, M. (2019). STACK is more than Maths – Development of Online Problems for Mechanics and Electrotechnics. *Contributions to the 1st International STACK conference 2018 in Fürth, Germany*. <https://doi.org/10.5281/zenodo.2577116>
- Mai, T., & Meyer, A. (2019). Sketching functions as a digital task with automated feedback. *Contributions to the 1st International STACK conference 2018 in Fürth, Germany*. <https://doi.org/10.5281/zenodo.2582427>
- Weigel, M., Hübl, R., Derr, K., & Podgayetskaya, T. (2019). STACK-Aufgaben im formativen eAssessment: Einsatzmöglichkeiten des Feedbacks. *Contributions to the 1st International STACK conference 2018 in Fürth, Germany*. <https://doi.org/10.5281/zenodo.2563656>
- Zerva, K. (2019). Developing STACK assessments in Edinburgh, 2017–2019. *Contributions to the 1st International STACK conference 2018 in Fürth, Germany*. <https://doi.org/10.5281/zenodo.2585816>

8 Selected publications from the fourth STACK conference, 2022

- Alarfaj, M., & Sangwin, C. J. (2022). Updating STACK potential response trees based on separated concerns. *International Journal of Emerging Technologies in Learning (iJET)*, 17, 94–102. <https://doi.org/10.3991/ijet.v17i23.35929>
- Guido Pinkernell, G., Diego-Mantecon, J. M., Lavicza, Z., & Sangwin, C. (2023). Authomath: Combining the strengths of stack and geogebra for school and academic mathematics. *International Journal of Emerging Technologies in Learning*, 18(3). <https://doi.org/10.3991/ijet.v18i03.36535>
- Hooper, C., & Jones, I. (2023). Conceptual Statistical Assessment Using JSXGraph. *International Journal of Emerging Technologies in Learning*, 18(1), 269–278. <https://doi.org/10.3991/ijet.v18i01.36529>

- Sangwin, C. J. (2023). Running an online mathematics examination with STACK. *International Journal of Emerging Technologies in Learning*, 18(3). <https://doi.org/10.3991/ijet.v18i03.35789>
- Zerva, K., & Sangwin, C. J. (2022). Rejuvenating the HELM workbooks as online STACK quizzes in 2020. *International Journal of Emerging Technologies in Learning (iJET)*, 17, 69–88. <https://doi.org/10.3991/ijet.v17i23.35923>

9 Legacy reports about STACK use

- O’Hagan, S., Sangwin, C. J., & Zerva, K. (2022). *A report on the use of the STACK service in Edinburgh, 2021–22* (tech. rep.). University of Edinburgh. Edinburgh, UK.
- Sangwin, C. J. (2010). *Who uses STACK? a report on the use of the STACK CAA system* (tech. rep.). The Maths, Stats and OR Network, School of Mathematics, The University of Birmingham.
- Sangwin, C. J. (2015). *Who uses STACK? a report on the use of the STACK CAA system* (tech. rep.). Loughborough University. Loughborough, UK.
- Sangwin, C. ., & Hermans, D. F. M. (2013). A report on the use of STACK in mathematics at Birmingham 2012–2013. *Community for Undergraduate Learning in the Mathematical Sciences*, 8, 16–30.
- Sporring, M., & Sangwin, C. J. (2019). *Who uses STACK? a report on the use of the STACK CAA system, August 2019* (tech. rep.). University of Edinburgh. Edinburgh, UK.

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